

ECS Washing machine

Stanley Lin Sheng Yang (p2120795), 30 June 2023
Wai Yan Aung (p2112422), Melinda

Lecturer name: Mr Ilyas

Overall system description

Introduction

Welcome to the design for a customizable washing machine! This design aims to provide users with the flexibility to select the type of material to be washed and choose from three different washing cycles: Rinse, Wash, and Spin. By allowing users to tailor the washing process based on their specific needs, this design seeks to enhance convenience and optimize cleaning performance.

Problems

- **Material Selection:** One of the challenges in this design is ensuring that the user can easily and accurately select the type of material they want to wash. The interface should provide clear options and guidance to assist the user in making the appropriate selection.
- **Cycle Time Variation:** Adjusting the cycle time based on the material chosen requires careful consideration. The design should account for the different cleaning requirements of various materials and provide optimal cycle durations for each. This can be achieved through proper research and testing to ensure effective cleaning without causing damage or excessive wear.
- **Washing Motion:** Implementing the three different washing motions—Rinse (back and forth motion), Wash (slow rotary motion), and Spin (fast rotary motion)—poses a technical challenge. The design needs to incorporate mechanisms that enable smooth and controlled motion transitions to deliver the desired washing effect for each cycle.
- **Safety and Reliability:** Ensuring the safety of the user and the washing machine is of utmost importance. The design must incorporate safety features such as automatic shut-off in case of any malfunction or irregularities. Additionally, the system should be reliable, durable, and resistant to potential failures during operation.
- **User Experience:** The design should prioritize user-friendliness, providing a straightforward and intuitive interface for easy operation. Clear instructions and feedback mechanisms should be incorporated to guide the user through the washing process. The inclusion of a simple melody upon cycle completion can enhance the

overall user experience, providing a satisfying and enjoyable interaction with the washing machine.

1) Product Requirements

Goals

- Provide the user with the ability to select the type of material to be washed.
- Allow the user to choose from three washing cycles: Rinse, Wash, and Spin.
- Vary the cycle time based on the selected material.
- Play a simple melody when the washing cycle completes.

Constraints

- The design should be user-friendly and intuitive to operate.
- The system should be able to handle different types of materials effectively.
- The design should ensure the safety of the user and the washing machine.
- The washing machine should be efficient and reliable in its performance.

2) System Analysis

2.1) Subsystem Identification

- Keypad
- LCD
- Stepper Motor
- Speaker

3) User Manual

The panel below the control panel for the washing machine. The control panel consists of open/close hatch button, material select button, start/continue button, pause button.



4) User Guide

1. On start, after “Welcome” Message is displayed, press “4” to open the hatch of the washing machine, user can now add in laundry into the washing machine.

Washing Machine

WELCOME

OPEN HATCH
BUTTON "4"



2. Press "5" to close hatch of the washing machine.

Washing Machine

"HATCH OPENED"
Please Put In Your Laundry

CLOSE HATCH
BUTTON "5"



3. Press either "1", "2", or "3" to select material for the type of material to be washed.

Washing Machine

"HATCH CLOSED"
Please Select Material Type
for Your Laundry

WOOL
BUTTON "1"

COTTON
BUTTON "2"

POLYESTER
BUTTON "3"



4. After selecting the material press “8” to begin the cleaning process.

Washing Machine

"MATERIAL SELECTED"
Press Start to Clean
Clothes

START/CONTINUE
BUTTON "8"



5. You should see 3 cycles of the cleaning process. Rinse Cycle (Motor moves back and forth), Wash Cycle (Motor moves in slow rotary motion), and finally Spin Cycle (Motor moves in fast rotary motion)

Washing Machine

"CLEANING CLOTHES NOW"

PAUSE
BUTTON "9"



6. Once the 3 cycles are completed, “cleaning done message will be displayed”, user can now press 4 to take out the cleaned laundry.

Washing Machine

"CLOTHES DONE CLEANING"

OPEN HATCH
BUTTON "4"

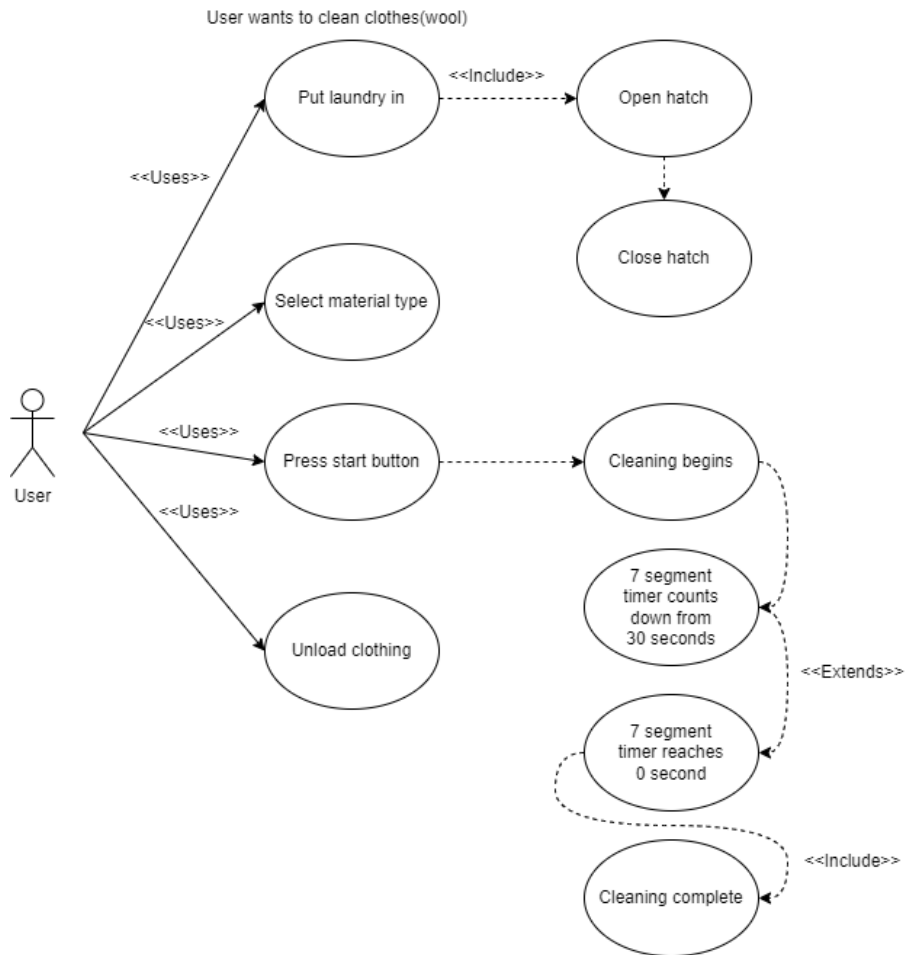


7. User can choose to restart the program.

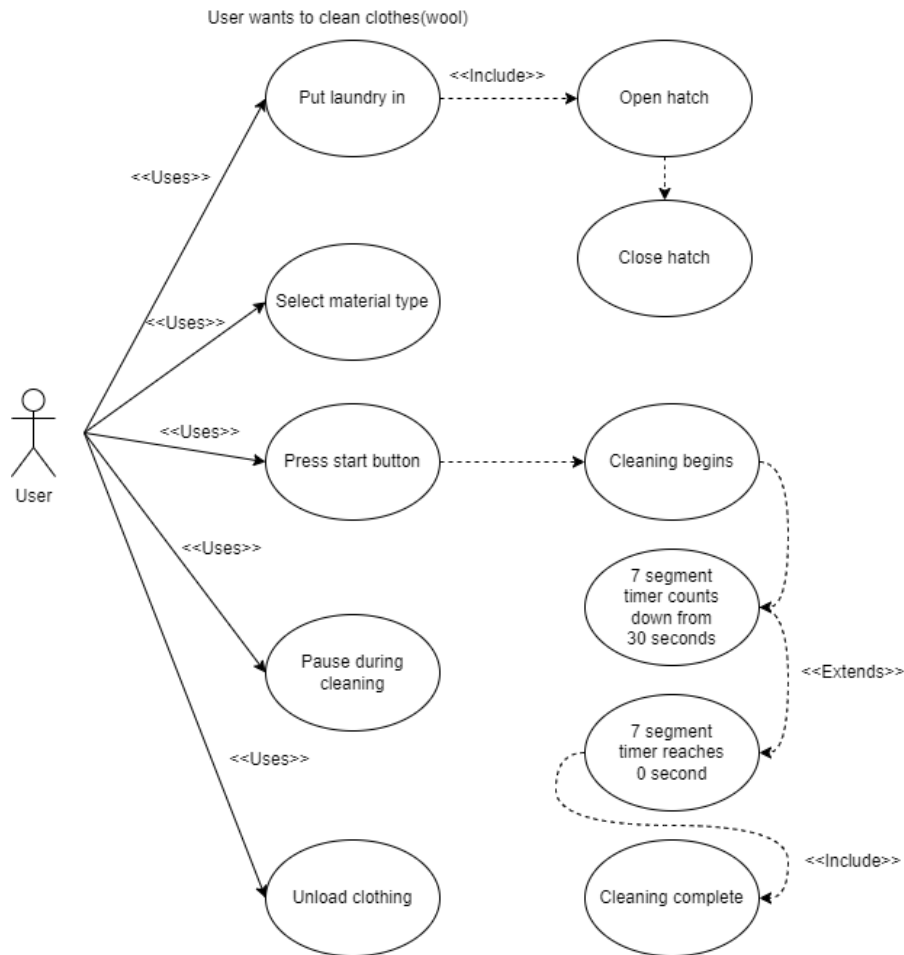
3) System Design

3.1) Use Case Diagrams

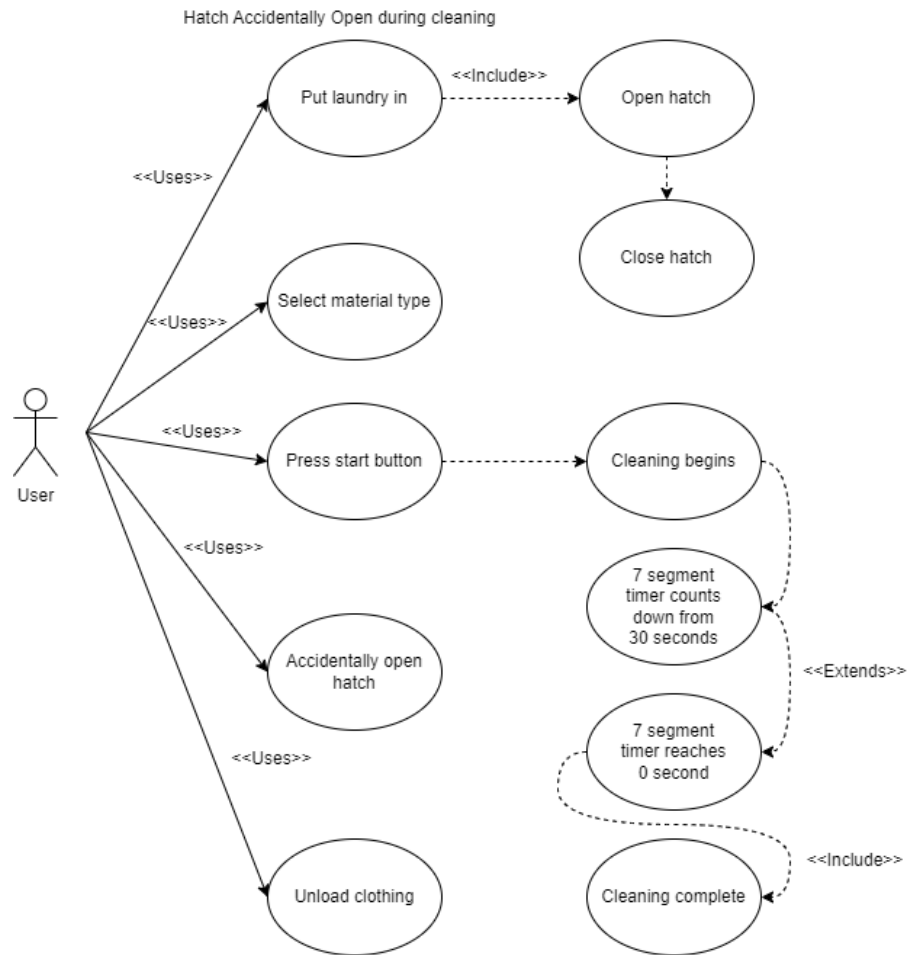
3.1.1) Normal Use Case Diagram



3.1.2) Pause during cleaning Use Case Diagram

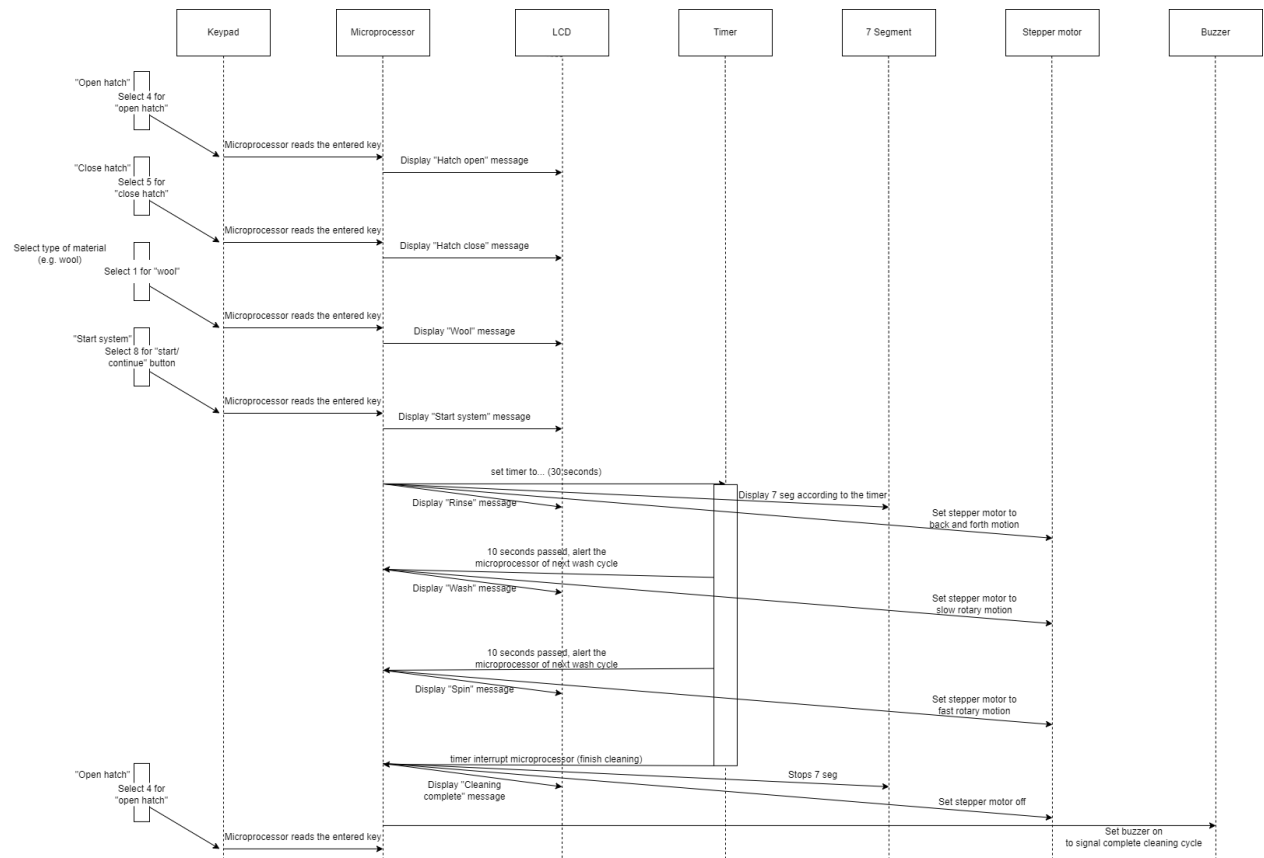


3.1.3) Accidentally Opened hatch Case Diagram

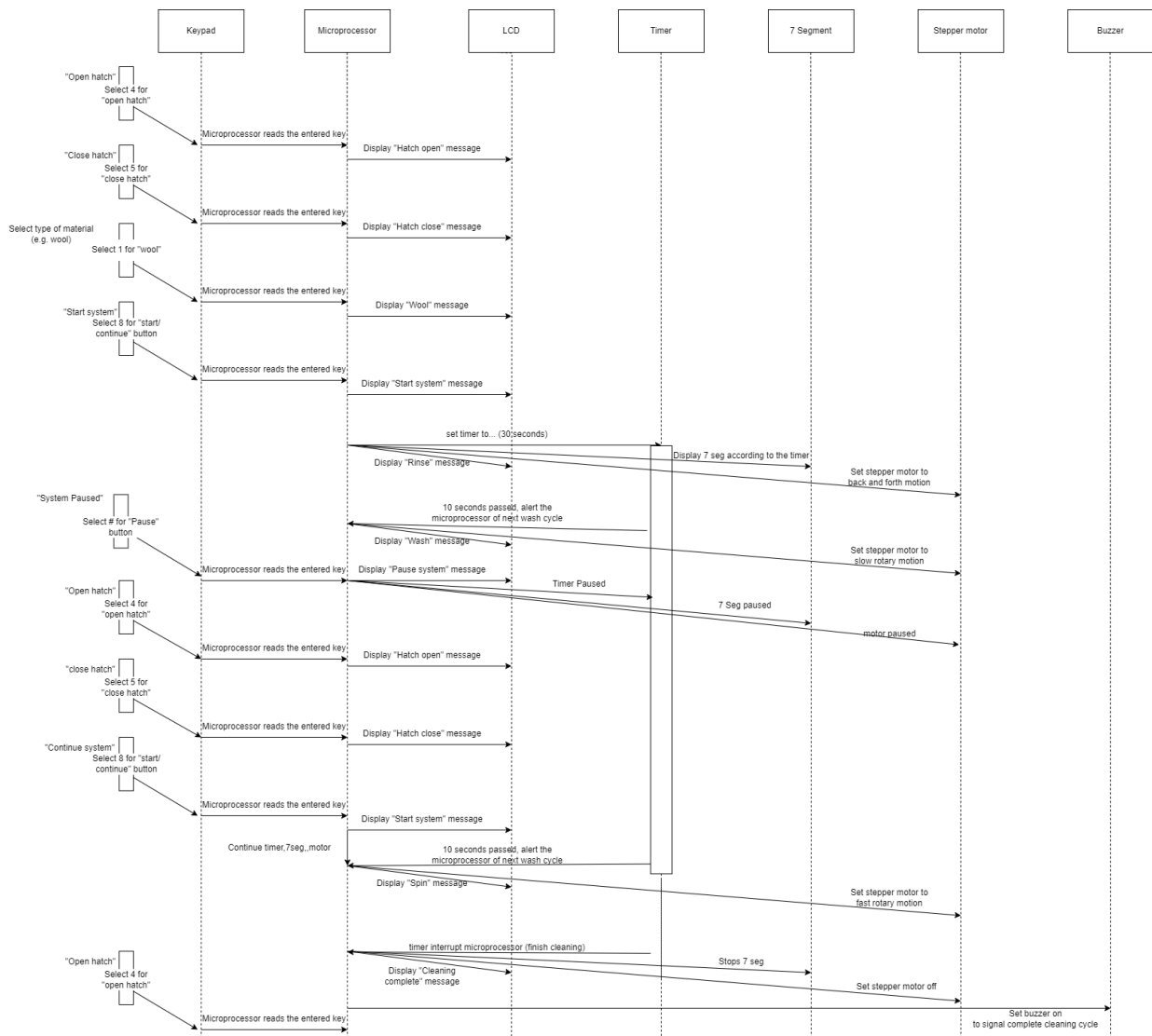


3.2) Interaction Diagrams

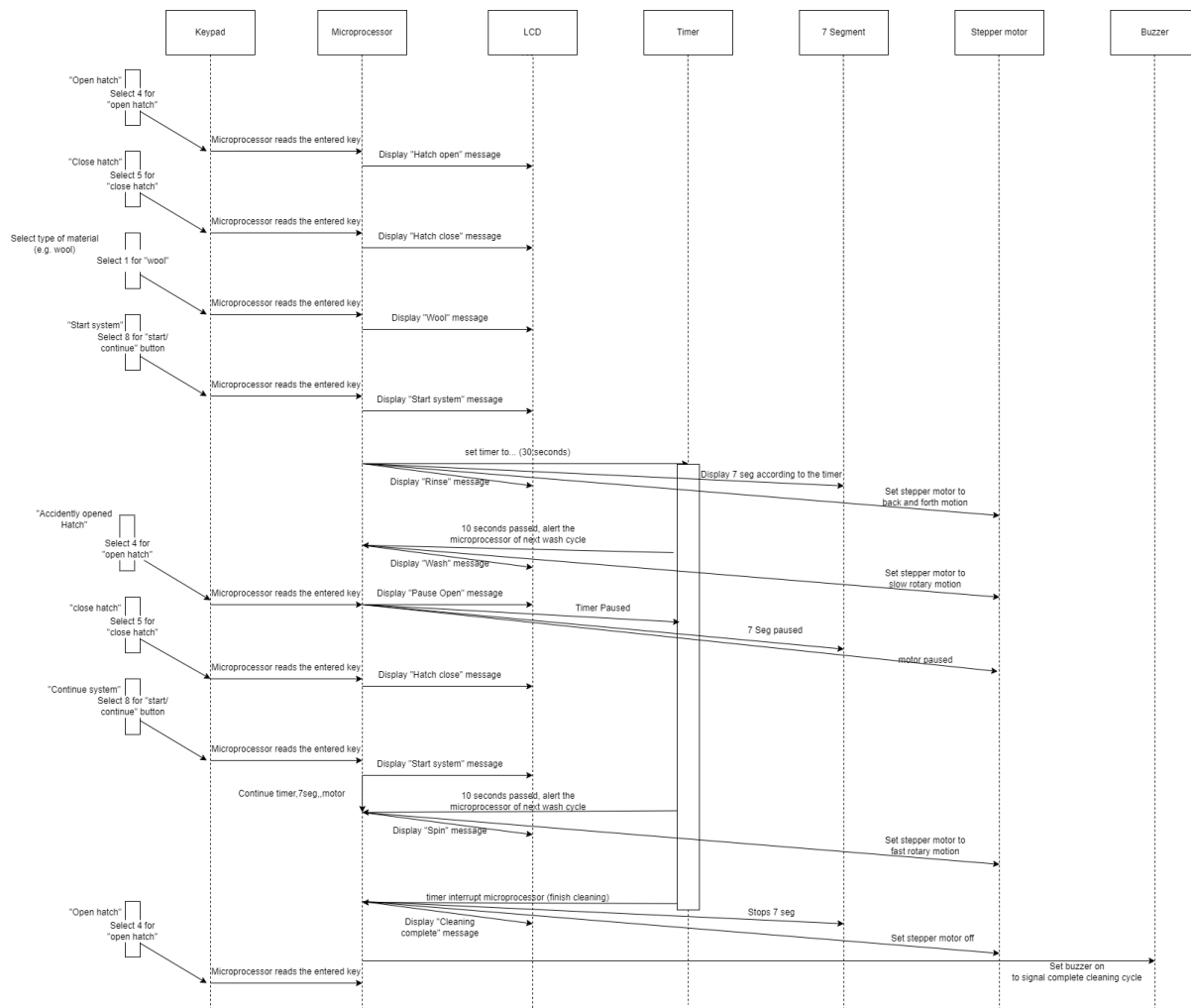
3.2.1) Normal Interaction Diagram



3.2.2) Adding Clothes Interaction Diagram



3.2.3) Accident Interaction Diagram



Work assigned for each student

Report

- Stanley Lin (33.33%)
- Wai Yan Aung (33.33%)
- Melinda (33.33%)

Coding

- Wai Yan Aung (50%)
- Stanley Lin (50%)

